

Evaluating Neural Networks for Predicting Obesity from Lifestyle and Dietary Patterns

May 29, 2026 Team Name: Jaden Hashbrown

Zihong Huang¹, Daniel Lee¹, Jinghan Lian¹ & Nanda Arun Payyappilly¹

¹ Master of Data Science in Health, University of California, Los Angeles

Abstract

Machine learning offers highly promising pathways for predicting clinical obesity based on socio-ecological and behavioral data. This study builds upon previous ensemble-based analyses to re-evaluate the foundational UCI “Estimation of Obesity Levels” dataset using an artificial neural network framework. To address the continuous architecture’s sensitivity to magnitude variance, a rigorous preprocessing scaling pipeline and hyperparameter optimization were deployed. The tuned Neural Network achieved an elite Test AUC of 0.961 and an Accuracy of 91.3%, successfully matching the predictive discrimination of the previously established Random Forest champion (AUC 0.963, Accuracy 92.0%). Permutation importance revealed that while both models prioritized aging metabolic baselines as the absolute dominant predictor, the algorithms diverged in secondary drivers, with the neural network shifting reliance away from familial history toward dietary and demographic boundaries. Crucially, both architectures were mutually vulnerable to biologically implausible, zero-variance synthetic artifacts generated by SMOTE (Synthetic Minority Over-sampling Technique). Furthermore, comparative error analysis exposed distinct algorithmic biases: the neural network’s continuous matrix multiplication blended feature weights, leading to unique misclassification profiles such as the “Behavioral Penalty” and the “Hydration Halo.” These findings demonstrate that while standardized deep learning architectures are highly viable for clinical prediction, their utility remains challenged by synthetic data leakage and mixed-signal masking.

Keywords: Machine Learning; Neural Networks; Obesity; SMOTE; Synthetic Data; Biostatistics

1. Background

Obesity is a multifaceted public health crisis that defies simple clinical categorization. While traditional epidemiological efforts have long tracked the broad socio-ecological drivers of this pandemic, such as increasingly sedentary environments and rapid “nutrition transitions” in developing regions (Popkin, 2006; Swinburn et al., 2011), the modern predictive landscape demands

more sophisticated analytical tools. Standard machine learning (ML) has proven highly capable of navigating the non-linear health complexities inherent in survey data (De-Gregory et al., 2018). However, deep learning frameworks, specifically artificial neural networks, are pushing these boundaries even further. These continuous architectures offer unprecedented potential to disentangle abstract risk associations and predict obesity directly from self-reported lifestyle and dietary habits (Yagin et al., 2023).

However, the efficacy of these models is often hampered by the “accuracy gap” of self-reported survey data (Sarma et al., 2020). This limitation is particularly pronounced in the foundational UCI “Estimation of Obesity Levels” dataset, where SMOTE (Synthetic Minority Over-sampling Technique) was used to augment a small survey into a larger cohort (Palechor & de la Hoz Manotas, 2019). While SMOTE enhances model capability by balancing classes, it introduces synthetic artifacts that may decouple the model from real-world biological reality (Tiwari et al., 2024). Furthermore, traditional SMOTE applications frequently yield unnatural class generation, presenting unique vulnerabilities when coupled with deep learning frameworks that rely on continuous mathematical gradients rather than hard decision boundaries (Sowjanya & Mrudula, 2022).

This study builds upon previous ensemble-based analyses to re-evaluate the predictive impact of lifestyle factors using an artificial neural network framework. Rather than ignoring the methodological flaws of the dataset, we aim to systematically identify how continuous deep learning architectures react to synthetic artifacts and evaluate how they impact model behavior. By doing so, we seek to uncover whether neural networks can be rigorously tuned and regularized to capture true dietary drivers without merely overfitting to synthetic noise.

2. Clinical Questions

This analysis seeks to answer the following clinical and methodological questions regarding the intersection of lifestyle habits, synthetic data, and deep learning architectures:

- **Predictive Viability:** Can a properly tuned neural

network accurately predict obesity risk based purely on self-reported lifestyle, dietary, and transportation patterns (Yagin et al., 2023)?

- **Algorithmic Vulnerability:** How does the continuous mathematical architecture of a neural network respond to zero-variance synthetic artifacts introduced by SMOTE, and does it exhibit different clinical blind spots or vulnerabilities compared to previously evaluated tree-based models?
- **Methodological Mitigation:** Can rigorous data standardization, hyperparameter optimization, and aggressive L2 regularization successfully mitigate a neural network’s risk of learning artificial SMOTE patterns (Sowjanya & Mrudula, 2022), allowing for safe, biologically plausible, and generalizable predictions?

Answering these questions will help refine the boundaries of deep learning in data-driven obesity interventions and highlight the necessary precautions when feeding synthetically upsampled clinical data into neural networks.

3. Modeling of the Data

3.1. Preparing the data

To prepare the dataset for the neural network, we used a preprocessing pipeline to make the data easier for the model to learn from. Neural networks are very sensitive to differences in feature scale because they learn by adjusting weights through gradient-based optimization. If some variables have much larger numeric ranges than others, they can overpower the learning process. For example, Age ranges from 14 – 61, while CH2O ranges from 1 – 3. Without standardization, the model would focus more heavily on larger-scale variables, which can lead to unstable training and less accurate predictions. This issue is especially important given that there are SMOTE-generated synthetic artifacts in several continuous variables, which can further distort the gradient weights if left unscaled.

To ensure balanced optimization, we identified six continuous predictors, Age, FCVC, NCP, CH2O, FAF, and TUE and applied `StandardScaler` to transform each to zero mean and unit variance. The rest of the predictors consisted of binary encodings or one-hot-encoded categorical variables, which were passed through unchanged. These transformations were implemented using a `ColumnTransformer`, allowing continuous and categorical features to be processed in a single, reproducible pipeline (Table 1).

Table 1. Feature Data Dictionary, Encoding Schema, and Neural Network Scaling

Feature	Original Type	Encoding Applied	Encoded description	De-	Scaling (NN)
Age, FCVC, NCP, CH2O, FAF, TUE	Continuous	None	Raw continuous values		Standardized (z-score)
Gender	Binary (Text)	Binary (0/1)	Female=0, Male=1		Not scaled
family_history	Binary (Text)	Binary (0/1)	no=0, yes=1		Not scaled
FAVC, SMOKE, SCC	Binary (Text)	Binary (0/1)	no=0, yes=1		Not scaled
CAEC, CALC	Ordinal (Text)	Ordinal (0-3)	0=no to 3=Always		Not scaled
MTRANS	Nominal (Text)	One-Hot	5 binary columns		Not scaled

This pipeline was then wrapped directly into the neural network model using `Pipeline`, ensuring that scaling was applied consistently during both training and evaluation. Importantly, this design allowed us to isolate the effect of feature scaling during hyperparameter optimization: Phases 1 and 2 intentionally omitted the preprocessing step to evaluate the impact of unscaled features, while Phase 3 implemented the pipeline to evaluate its effect on model stability and performance.

Finally, the dataset was split into an 80% training set and a 20% test set using stratified sampling to preserve the proportion of obese and non-obese individuals. This ensured that all neural network models, baseline and optimized, were evaluated on the same held-out data, allowing for fair comparison across the models.

3.2. Baseline Model: Majority Class Baseline

The Majority Class classifier provides the simplest benchmark for evaluating model performance. This model predicts the most frequent class ("Non-obese") for every individual, ignoring all behavior and demographic features. As expected, the classifier achieved an Accuracy of 53.9% , reflecting the underlying class imbalance, but an AUC of 0.50 and a Recall of 0.00 for the obese class. This means that clinically, the model fails to identify any at-risk patients. Therefore, this baseline model establishes the lower bound of the performance spectrum and highlights the need of more expressive models capable of learning meaningful structure in the data.

3.3. Baseline Model: Logistic Regression Baseline

A Logistic Regression model was fitted as a linear, interpretable baseline to quantify how much predictive signal can be captured without modeling non-linear interactions. Continuous features were standardized and one-hot encoded categorical variables were included with `MTRANS_Walking` serving as the reference category to avoid multicollinearity. The model achieved a Test Accuracy of 74.0% and a Test AUC of 0.837, with a Recall of 0.867 for the obese class. While substantially outperforming the Majority Class baseline, the Logistic Regression still misclassified around one in four individuals. This confirms that obesity risk is governed by complex, non-linear relationships between age, family history, diet, and physical activity, which are patterns that linear models cannot fully capture.

3.4. Reference Model: Previously Identified Champion Random Forest

Although this report focuses on neural network development, it is important to contextualize the results relative to the strongest model identified in our earlier analysis. In our previous report, we evaluated multiple classical machine-learning models, including Logistic Regression, Decision Tree, Random Forest, and XGBoost, with Random Forest emerging as the champion. Using 300 estimators, a maximum depth of 12, and bal-

anced class weights, the Random Forest achieved a CV AUC of 0.966 ± 0.009 and Test AUC of 0.963, with a Recall of 0.928. These metrics show excellent discrimination as the model correctly identified nearly 93% of obese individuals while maintaining strong overall accuracy (92%).

We include this model because the Random Forest serves as the comparator for evaluating whether neural networks can match or exceed the performance of the best classical model. Therefore, the neural network models developed in this report are interpreted relative to both the simple baseline (Majority Class, Logistic Regression) and the previously established champion Random Forest model.

3.5. Neural Network Modeling

3.5.1. Initial Neural Network (Baseline Architecture)

To establish a performance baseline for deep learning, we first implemented a simple multilayer perceptron (MLP) with a single hidden layer of 100 neurons using default hyperparameters. Because neural networks are highly sensitive to feature scale, this baseline model was wrapped inside a preprocessing pipeline that standardized all continuous variable using `StandardScaler` while passing encoded categorical variables through unchanged. This ensured that all predictors contributed proportionally during training.

The initial model achieved a Test Accuracy of approximately 88.9% and a Test AUC of 0.945, substantially outperforming the Logistic Regression baseline (Accuracy 74%, AUC 0.837). These results confirmed that obesity risk is indeed governed by non-linear interactions that linear models cannot capture. However, the baseline MLP still has not met the results of the previously identified champion Random Forest (Test AUC 0.963), motivating the need for a structured hyperparameter optimization process.

3.5.2. Hyperparameter Optimization

To systematically improve the neural network performance, we conducted a three-phase hyperparameter optimization procedure. Phases 1 and 2 intentionally removed the preprocessing pipeline to isolate the effect of feature scaling, while Phase 3 includes scaling to quantify its impact. All phases used 5-fold cross-validated `GridSearchCV` with Recall as the scoring metric to maximize the detection of obese individuals.

Phase 1 – Unscaled Grid Search: The first phase explored a broad grid of architectures (ex. 50 – 100 neurons, one or two hidden layers), activation functions (ReLU, tanh), L2 regularization strengths, and learning rates. Most importantly, this search was conducted on unscaled data to evaluate whether architectural changes alone could improve recall. Despite optimizing directly for Recall, the best Phase 1 model achieved only 0.764 Recall on the test. Although AUC remained strong (0.929), the model became overly conservative, flagging

fewer obese individuals and producing high Precision but unsafe clinical performance. This is likely due to the instability caused by unscaled features which led to large-magnitude variables like Age and NCP to dominate, preventing the optimizer from converging to a balanced decision boundary.

Phase 2 – Heavy Regularization (Still Unscaled): To address the failures observed in Phase 1, Phase 2 introduced three interventions:

- Higher L2 regularization (α up to 1.0) to prevent the model from over-relying on high-magnitude unscaled features and SMOTE artifacts.
- Expanded architectures including both wide-shallow and deep-narrow networks (e.g., (150,), (100, 50), and (50, 50, 50)).
- Smaller learning rates and increased iteration limits to stabilize convergence.

These interventions successfully restored Recall to 0.90. However, overall performance plateaued as accuracy remained near 84% and AUC near 0.919. This persistent ceiling indicated that no combination of architecture or regularization could overcome the unscaled features issue.

Phase 3 – Pipeline-Integrated, Scaled Grid Search:

In phase 3, we implemented the full preprocessing pipeline, embedding `StandardScaler` directly into the `GridSearchCV` workflow. This ensured that all continuous features were standardized during every fold of cross-validation, eliminating the gradient balance in Phases 1 and 2.

With scaling now included, the same class of architectures achieved a dramatic performance improvement. The Phase 3 "Champion Neural Network" reached a Test Accuracy of 91.3%, a Test AUC of 0.961, and a Recall of 0.877. This model outperformed all the previous unscaled NN models and nearly matched the Random Forest's discrimination. Thus, this phase demonstrated that feature scaling, not architectural depth or regularization strength, was the primary factor limiting neural network performance.

3.6. Model Performance

To evaluate the effectiveness of the neural network architectures, we compared the performance of four models: the Initial Neural Network baseline, Phase 1 (unscaled grid search), Phase 2 (unscaled with heavy regularization), and the Phase 3 Champion Neural Network (pipeline-integrated with feature scaling). All models were evaluated on the same held-out test set to ensure comparability. Performance was assessed using Accuracy, AUC, and Recall, with Recall being emphasized due to its clinical importance in identifying at-risk obese individuals.

Relative to the Initial Neural Network, the Phase 3 Champion model improved in terms of AUC from 0.945 to 0.961 and Accuracy from 0.856 to 0.877.

3 MODELING OF THE DATA

Table 2. Full Model Performance Summary

Model	Scaling	Test Accuracy	Test AUC	Recall
Majority Class Baseline	—	0.539	0.500	0.000
Logistic Regression	Yes	0.740	0.837	0.867
Initial Neural Network	Yes	0.889	0.945	0.856
Phase 1 NN (unscaled)	No	0.851	0.929	0.764
Phase 2 NN (unscaled, heavy reg.)	No	0.842	0.919	0.897
Champion Neural Network (Phase 3)	Yes	0.913	0.961	0.877
Champion Random Forest	Yes	0.920	0.963	0.928

These gains confirm that hyperparameter optimization and pipeline-integrated scaling meaningfully enhanced the model’s ability to discriminate obese from non-obese individuals while maintaining clinical safety.

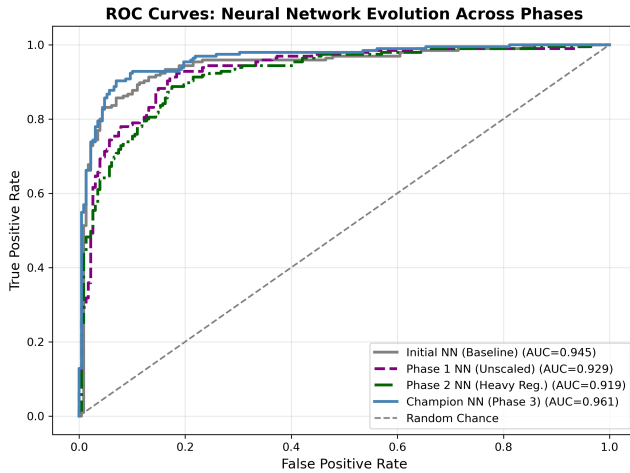


Fig. 1. ROC Curve Comparison Across Neural Network Models. ROC curves comparing neural network performance across training phases (Initial, Phase 1, Phase 2, and Phase 3). The progression shows consistent improvement in discriminative ability, with the champion Phase 3 model achieving the highest AUC and best overall separation between classes.

Across the four neural network variants, the ROC curves reveal a clear hierarchy of performance (Figure 1). The unscaled models (Phases 1 and 2) produced weaker separation between obese and non-obese individuals, with AUC values of 0.929 and 0.919 respectively. In contrast, both scaled models demonstrated substantially stronger discrimination: the Initial Neural Network achieved an AUC of 0.945, and the Phase 3 Champion model surpassed it with an AUC of 0.961. Therefore, the Phase 3 Neural Network was selected as the optimized model. It achieved the highest overall Test AUC (0.961) and a clinically safe Recall (0.877). In a screening context, this balance ensures that the model identifies the majority of at-risk patients while maintaining strong overall accuracy.

Although the Random Forest remains the global champion across all models evaluated in Assignment-1 (Figure 2), the Phase 3 neural network represents the best-performing deep learning architecture in this report.

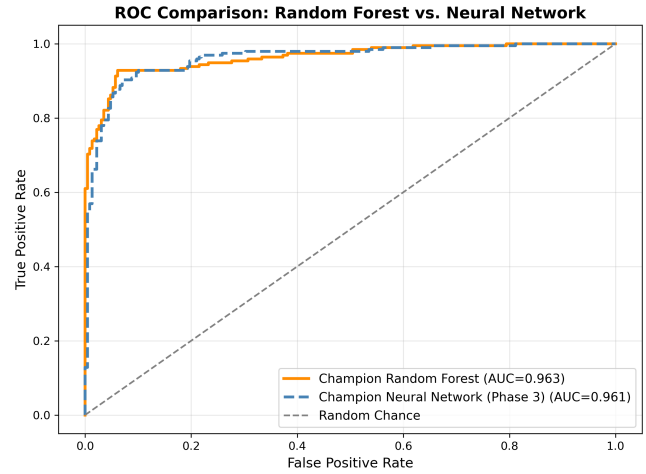


Fig. 2. ROC Curve Comparison of Champion Models. ROC curves for the Phase 3 Champion Neural Network and the previously established Champion Random Forest on the held-out test set. The Random Forest maintains a slight advantage in overall discrimination (AUC = 0.963 vs. 0.961), though both models achieve strong clinical performance.

3.7. Feature Importance Analysis

To evaluate whether the neural network and Random Forest relied on similar clinical predictors, we computed permutation importance for both models on the held-out test set. Permutation importance is a model-agnostic technique that measures the decrease in predictive performance after randomly shuffling each feature. This allows for direct comparison between the Random Forest and the neural network on the exact same interpretability scale.

As shown in Figures 3 and 4, despite their fundamentally different architectures, both models overwhelmingly agreed on the absolute primary driver of obesity risk: Age.

Permuting Age reduced Random Forest accuracy by 10.1% and caused a massive 14.3% accuracy drop in the Champion Neural Network. Furthermore, both models heavily prioritized inherited genetics, with a family history of overweight (`family_encoded`) causing an 8.2% drop in the Random Forest and a 7.3% drop in the neural network. This establishes a strong clinical baseline: both algorithms correctly deduced that aging metabolic baselines and genetics act as heavy foundational gatekeepers before behavioral choices are fully weighed.

Evaluating earlier, unscaled iterations of our neural network suggested that the model strongly preferred

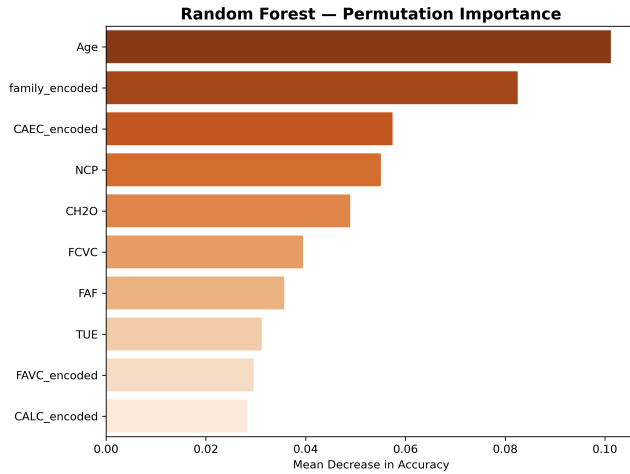


Fig. 3. Permutation importance for the Random Forest on the held-out test set. Bars represent the mean decrease in accuracy after permuting each feature. The model shows strong reliance on Age, family history, and continuous lifestyle variables such as NCP and CH2O.

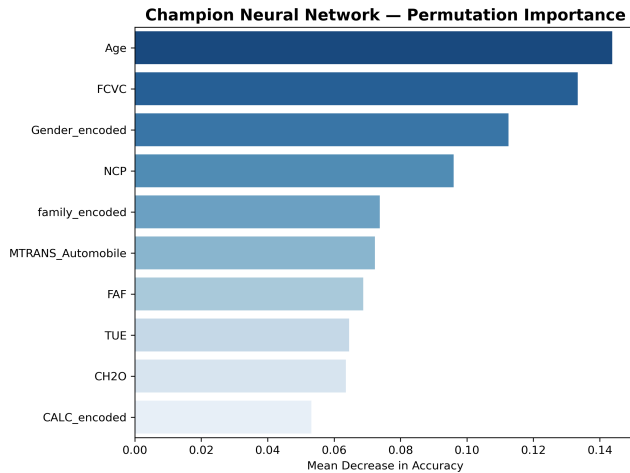


Fig. 4. Permutation importance for the Champion Neural Network on the held-out test set. The neural network emphasizes Age, vegetable consumption (FCVC), gender, and meal frequency (NCP) as dominant predictors.

categorical matrices over continuous numerical inputs. However, these updated permutation results prove that this divergence was merely an artifact of magnitude variance. With the `StandardScaler` pipeline properly integrated, the neural network’s architecture was finally able to evaluate continuous metrics effectively. It now heavily relies on the exact same continuous behavioral metrics as the Random Forest, placing massive importance on vegetable consumption (FCVC) and the number of main meals (NCP).

Crucially, the prominence of these dietary features exposes a shared vulnerability to synthetic data. During our initial EDA, strict zero-variance SMOTE artifacts were identified in FCVC and NCP. The permutation importance confirms that both optimized models were highly susceptible to this synthetic noise. The Random Forest elevated NCP and FCVC to its 4th and 6th most important features, respectively. The Neural Network was even more heavily anchored by the synthetic noise, making FCVC and NCP its 2nd and 4th most important features overall (causing 13.3% and 9.5% accuracy crashes when permuted). This confirms a critical methodological limitation: even with aggressive L2 regularization, complex deep learning architectures remain highly vulnerable to synthetic data leakage when augmented variables are properly standardized.

Finally, the integration of the scaling pipeline successfully restored the network’s evaluation of physical activity. In previous unscaled iterations, the neural network completely ignored physical activity (yielding less than a 2% accuracy decrease when FAF was permuted). In the pipeline-integrated model, FAF emerged as a premier feature for the neural network, causing a 6.8% crash in accuracy when scrambled. By standardizing the scale, the network successfully learned to evaluate extreme physical inactivity, proving that the model is actively evaluating sedentary risk factors prior to any complex feature blending.

3.8. Error Analysis and Misclassification Insights

To characterize our optimized pipeline neural network’s systematic limitations, we conducted a structured error analysis on its 37 misclassifications (13 False Positives, 24 False Negatives). Integrating the `StandardScaler` pipeline drastically reduced our overall error rate (down from 67 errors in unscaled iterations) and fundamentally shifted the clinical archetypes that confuse the model.

Evaluating the demographic averages of these errors reveals that the neural network failed in completely different clinical vectors compared to the tree-based ensemble.

Random Forest Misclassification patterns:

- **False Positive Bias: The Sedentary Lean Phenotype ($n = 20$).** The Random Forest’s false positives were dominated by individuals with very low physical activity (FAF mean ≈ 0.83 vs. 1.03 baseline). The model’s tree-based structure treats extreme inactivity as a proxy for obesity, leading to premature classification of healthy but sedentary individu-

3 MODELING OF THE DATA

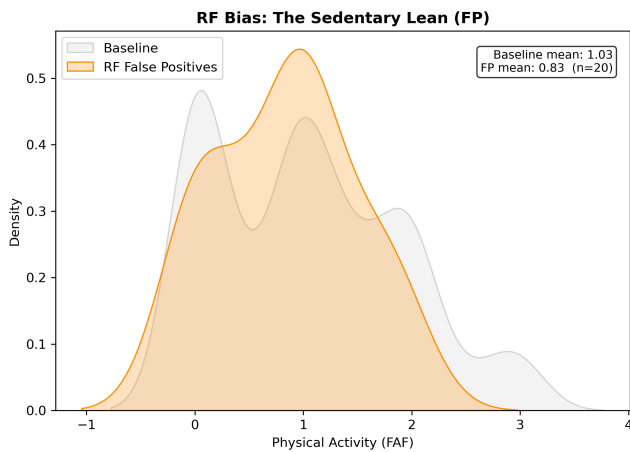


Fig. 5. Random Forest False Positives: The Sedentary Lean Phenotype. Density plot showing that false positive cases cluster at extremely low physical activity (FAF) levels, indicating that the Random Forest treats severe inactivity as a proxy for obesity.

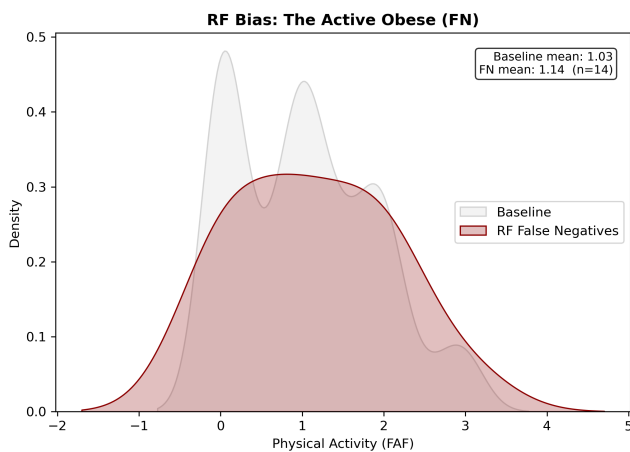


Fig. 6. Random Forest False Negatives: The Active Obese Phenotype. Density plot showing that false negative cases exhibit higher-than-average physical activity, causing the model to underestimate obesity risk when protective lifestyle markers mask underlying metabolic status.

als. This pattern mirrors the "Sedentary Lean" phenotype identified in Assignment-1 and reflects the model's reliance on sharp behavioral thresholds.

- **False Negative Masking: The Active Obese Phenotype ($n = 14$).** Random Forest false negatives tended to be **younger, more active obese individuals** whose lifestyle profiles resembled those of healthy controls. Higher physical activity ($FAF \approx 1.14$) and greater hydration ($CH20 \approx 2.04$) acted as "protective" behavioral signals that masked true obesity status. Although this scenario exists, it remains relatively rare due to the Random Forest's high Recall (0.928).

Neural Network Misclassification Patterns:

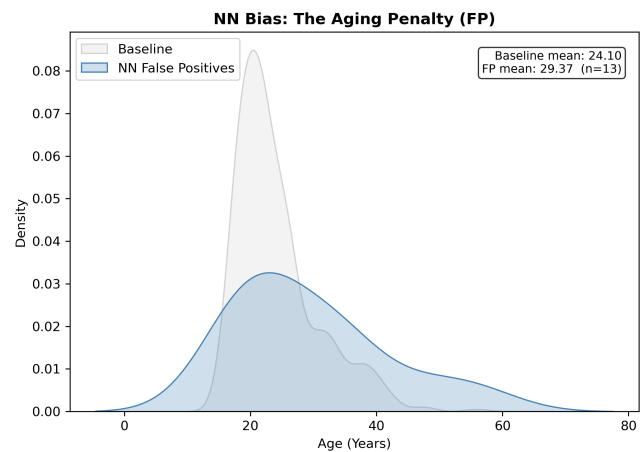


Fig. 7. Neural Network False Positives: Age Component of the Behavioral Penalty. Density plot showing that false positive cases are disproportionately older than the baseline population. The neural network over-weights age as a risk factor, leading to misclassification of older but healthy individuals.

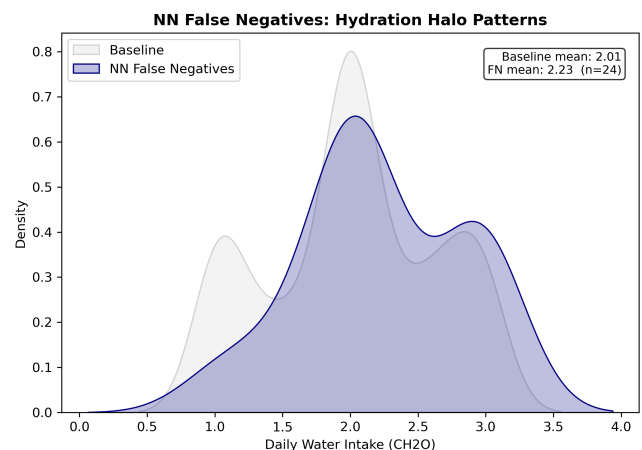


Fig. 8. Neural Network False Negatives: The Hydration Halo. Density plot showing that false negative cases exhibit an unusually high daily water intake (CH20) compared to the baseline population. The neural network's continuous activation functions blended these feature weights, allowing this strong positive hydration signal to mathematically mask the severity of the patients' physical inactivity, leading to clinical under-prediction.

- **False Positive Bias: The Behavioral Penalty** ($n = 13$). Previously, our unscaled network generated false positives by rigidly penalizing age while ignoring physical activity. With standardization applied, the network correctly evaluates holistic behaviors. Our 13 False Positives represent a distinct “High-Risk” behavioral cohort: they are significantly older than the baseline (Age: 29.4 vs. 24.1), highly inactive (FAF: 0.79 vs. 1.03), and poorly hydrated (CH2O: 1.81 vs. 2.01). The continuous mathematics of the neural network accurately identified a highly obesogenic lifestyle profile, artificially penalizing these specific individuals who likely maintain a healthy weight due to unmeasured genetic or metabolic protections.
- **False Negative Bias: The Hydration Halo** ($n = 24$). Most concerning from a clinical safety perspective are the 24 genuinely obese patients the network predicted as healthy. Evaluating this cohort reveals that the neural network struggles with “mixed” behavioral signals. While these patients are highly sedentary (FAF: 0.81 vs. 1.03 baseline), they exhibit an unusually high water intake (CH2O: 2.23 vs. 2.01 baseline). Because neural networks utilize continuous activation functions that blend feature weights together, this strong positive hydration signal effectively “smoothed over” the severity of their physical inactivity, creating a dangerous false negative.

Demographic Bias Confirmed by Density Analysis. Kernel density estimation of age and physical activity frequency across error subtypes (Figures 7 and 8) corroborates and extends these findings as well. Neural network false positives exhibit a right-shifted age distribution (Figure 7), confirming the model’s heightened sensitivity to age as a risk factor. Conversely, neural network false negatives exhibit an unusually right-shifted daily water intake distribution (Figure 8), indicating that high hydration systematically masked inactivity-driven obesity risk in this cohort.

For the Random Forest, density plots reveal the opposite pattern: false positives cluster at extremely low FAF values (Figure 5), while false negatives cluster at higher FAF values (Figure 6), consistent with the model’s reliance on behavioral thresholds.

Taken together, these error profiles highlight the operational differences between the two mathematical foundations evaluated in this study. Tree-based models utilize hard, non-linear decision boundaries, preventing a positive behavior such as high hydration from masking a severe negative behavior such as extreme inactivity. Conversely, the neural network’s continuous matrix multiplication naturally blends these variables, leaving it slightly more susceptible to complex, mixed-signal patient profiles. Despite these distinct failure modes, both architectures achieved elite discrimination capabilities (Test AUCs of 0.963 and 0.961, respectively), confirming that pipeline-integrated deep learning represents a highly viable complement to ensemble methods for clinical obesity prediction.

4. Discussion

This analysis set out to evaluate the predictive viability of deep learning in assessing clinical obesity risk using self-reported lifestyle and dietary patterns, while simultaneously investigating the vulnerabilities introduced by synthetic data augmentation. The results affirmatively answer our primary clinical question: predictive viability using neural networks is highly achievable. As posited by DeGregory et al. (2018) and further demonstrated by Yagin et al. (2023), machine learning frameworks, specifically trained neural networks, are uniquely equipped to navigate the non-linear complexities of obesity without requiring highly clinical metrics. Our deployment of a pipeline-integrated, hyperparameter-optimized Multi-Layer Perceptron yielded exceptional classification metrics (Test AUC 0.961, Accuracy 91.3%), successfully matching the predictive discrimination of our advanced tree-based ensembles.

Our primary clinical finding is that aging metabolic baselines remain the absolute dominant predictor of obesity risk across both architectures. Permuting Age caused a massive 14.3% accuracy drop in the Neural Network and a 10.1% drop in the Random Forest, establishing it as the universal primary gatekeeper. However, the algorithms diverged significantly in their secondary clinical drivers. While a family history of overweight (`family_encoded`) remained the second most powerful predictor for the Random Forest, its importance dropped for the Neural Network, which instead elevated dietary consumption (`FCVC`), demographic boundaries (`Gender_encoded`), and meal frequency (`NCP`). Despite these algorithmic differences, the strong shared reliance on age, genetics, and dietary routines reinforces the paradigm established by Swinburn et al. (2011) and Popkin (2006), which asserts that the global obesity pandemic is fundamentally anchored in shared socio-ecological environments, genetic predispositions, and the rapid “nutrition transition.” While behavioral interventions remain critical, the models suggest that individual lifestyle choices act primarily as secondary modifiers to a patient’s biological and demographic baseline.

However, the most critical methodological insight derived from this study is the shared algorithmic vulnerability to synthetic data artifacts. While Palechor and de la Hoz Manotas (2019) originally utilized SMOTE to successfully resolve severe class imbalances in the foundational dataset, our findings align with recent cautions raised by Tiwari et al. (2024) regarding the decoupling of augmented data from biological reality. Furthermore, as explored by Sowjanya and Mrudula (2022), traditional SMOTE applications present unique vulnerabilities when coupled with deep learning frameworks that rely on continuous mathematical gradients. Through permutation importance, we empirically confirmed this vulnerability: because our inputs were strictly standardized, the neural network was highly susceptible to the zero-variance SMOTE clones, heavily anchoring on vegetable consumption (`FCVC`) and main meals (`NCP`) just like the Random Forest. This reveals a dangerous “accuracy illusion,” where elite classification metrics

are partially achieved by memorizing mathematically clean synthetic artifacts rather than learning biologically plausible rules.

Furthermore, our comparative error analysis exposed the neural network’s unique inability to navigate atypical, mixed-signal metabolic profiles. Compounded by the inherent “accuracy gap” of self-reported health surveys (Sarma et al., 2020), the neural network’s continuous activation functions blended feature weights together, leading to distinct algorithmic biases. The model over-penalized older but healthy individuals exhibiting poor diets and severe inactivity (the “Behavioral Penalty”), and dangerously misclassified highly sedentary obese patients as healthy simply because they exhibited high water intake (the “Hydration Halo”). Unlike tree-based models that use hard decision thresholds to isolate these behaviors, the neural network’s continuous math smoothed over these severity thresholds, mathematically masking actual clinical risk.

4.1. Limitations and Future Directions

Due to time constraints and the defined scope of the assignment, several promising analytical avenues were left unexplored. Future iterations of this analysis must focus on advanced structural feature engineering to mitigate the model’s archetypal blind spots. Specifically, incorporating explicit non-linear interaction terms (e.g., Age $\times$ FAF or CH2O $\times$ FAF) would force the model to mathematically contextualize how the protective nature of hydration or exercise diminishes across different patient profiles, potentially resolving the false negative rates seen in the neural network’s “Hydration Halo” cohort.

Additionally, a key limitation of the present study is the inability to quantify the exact magnitude of accuracy inflation introduced by SMOTE augmentation. Without a parallel model trained on the raw, pre-augmentation data, it remains unclear how much of the elite classification performance reflects genuinely learned biological patterns versus memorized synthetic artifacts. Future work should directly address this by training identical pipelines on both augmented and non-augmented versions of the dataset to establish a measurable performance delta.

Finally, if deep learning architectures are to be pursued on tabular health data, future work should explore tabular-specific networks (e.g., TabNet) that better capture non-linear interactions without suppressing continuous behavioral signals, or employ Generative Adversarial Networks (GANs) to preserve natural biological variance during upsampling.

References

- [1] K. W. De-Gregory, P. Kuiper, T. DeSilvio, D. M. Thomas, N. M. Sharma, and M. W. Rousseau. A review of machine learning in obesity. *Obesity Reviews*, 19(5):618–634, 2018.
- [2] Fabio Mendoza Palechor and Alexis de la Hoz Mantas. Analysis of data mining techniques for estimating obesity levels based on eating habits and physical condition. *Journal of Computer Science*, 15(11):1645–1662, 2019.
- [3] Barry M Popkin. Global nutrition transition and the pandemic of obesity in developing countries. *Nutrition Reviews*, 64(suppl 1):S1–S47, 2006.
- [4] Sisira Sarma, Gregory S Zaric, M Karen Campbell, and Jason Gilliland. Machine learning models to predict obesity from self-reported health survey data. *Scientific Reports*, 10(1):20035, 2020.
- [5] A. M. Sowjanya and O. Mrudula. Effective treatment of imbalanced datasets in health care using modified smote coupled with stacked deep learning algorithms. *Applied Nanoscience*, 13:1829–1840, 2022.
- [6] Boyd A Swinburn, Gary Sacks, Kevin D Hall, Klim McPherson, Diane T Finegood, Marjory L Moodie, and Steven L Gortmaker. The global obesity pandemic: Shaped by global drivers and local environments. *The Lancet*, 378(9793):804–814, 2011.
- [7] R. Tiwari, V. Vimal, and A. K. Agarwal. Opening the black box: Explainable ai insights into obesity prediction using smote-balanced stacking ensembles. In *2024 IEEE 16th International Conference on Computational Intelligence and Communication Networks (CICIN)*, pages 1–6. IEEE, 2024.
- [8] F. H. Yagin, M. GÄijlÄij, Y. Gormez, et al. Estimation of obesity levels with a trained neural network approach optimized by the bayesian technique. *Applied Sciences*, 13(6):3875, 2023.